

ASSIGNMENT

- 1 Prove that the number $30^{239} + 239^{30}$ is not prime.

Solution:

$$30^{239} + 239^{30} \equiv (-1)^{239} + 1 \equiv 0 \pmod{31}$$

Number divisible by 31 so not prime.

- 2 Let p be a prime number and suppose a and b are arbitrary integers. Prove that $(a + b)^p \equiv a^p + b^p \pmod{p}$

Solution:

$$(a + b)^p \equiv (a + b)^p \equiv a^p + b^p \pmod{p}$$

- 3 The sum of the numbers a, b and c is divisible by 30. Prove that $a^5 + b^5 + c^5$ is also divisible by 30.

Solution:

$$x^5 - x = (x-1)(x)(x+1)(x^2 + 1)$$

In three consecutive numbers, there is at least one number is multiple of 3 and one is multiple of 2

$$x = 5k, 5k+1, 5k+2, 5k+3, 5k+4$$

It can be easily observed that, for $x = 5k, 5k+1$ and $5k+4$, $(x-1)(x)(x+1)(x^2 + 1)$ divisible by 5.

For $x = 5k+2$ & $5k+3$, $x^2 + 1$ is divisible by 5.

$$\Rightarrow x^5 - x \text{ is divisible by 30.}$$

$$\Rightarrow a^5 + b^5 + c^5 - a - b - c \text{ is divisible by 30.}$$

Given $a + b + c$ is divisible by 30 therefore $a^5 + b^5 + c^5$ is also divisible by 30.

- 4 Let p and q be different primes. Prove that $p^q + q^p \equiv p + q \pmod{pq}$

Solution:

$$p^q \equiv p \pmod{q} \text{ \& } p^q \equiv 0 \pmod{p}$$

$$\Rightarrow p^q \equiv p \pmod{pq} \dots (i)$$

$$q^p \equiv q \pmod{p} \text{ \& } q^p \equiv 0 \pmod{q}$$

$$\Rightarrow q^p \equiv q \pmod{pq} \dots (ii)$$

Adding (i) & (ii)

$$p^q + q^p \equiv p + q \pmod{pq}$$

- 5 Let p and q be different primes, Prove that $\left[\frac{p^q + q^p}{pq} \right]$ is even if $p, q \neq 2$, where $[x]$ denotes the integral part of

x .

Solution:

From previous question $p^q + q^p \equiv p + q \pmod{pq}$

$$\Rightarrow p^q + q^p - p - q = kpq$$

p and q must be odd,

$$\Rightarrow (p^q + q^p - p - q) \text{ must be even}$$

$$\Rightarrow k \text{ is even number}$$

$$p^q + q^p = p + q + kpq$$

$$\frac{p^q + q^p}{pq} = \frac{p + q}{pq} + k$$

$\frac{p+q}{pq}$ is a fraction and k is integral part which is even

- 6 Let p be prime, and suppose p does not divide some number a . Prove that there exists a natural number b such that $ab \equiv 1 \pmod{p}$.

Solution:

Let us set $b = a^{p-2}$.

$$\text{Then } ab = a^{p-1} \equiv 1 \pmod{p}$$

- 7 Show that $\sum_{r=1}^{n-3} r(r)!$ is divisible by n if and only if n is prime.

Solution:

$$\sum_{r=1}^{n-3} r(r)!\equiv \sum_{r=1}^{n-3} (r+1-1)(r)!\equiv \sum_{r=1}^{n-3} (r+1)!-(r)!\equiv (n-2)!-1$$

ATQ if $(n-2)!-1$, divisible by n .

$$(n-2)!-1 \equiv -(n-1)!-1 \equiv 0 \pmod{n}$$

$$(n-1)! \equiv -1 \pmod{n} \Rightarrow n \text{ is prime by Wilson theorem.}$$

- 8 Let n be a natural number not divisible by 17. Prove that either $n^8 + 1$ or $n^8 - 1$ is divisible by 17.

Solution:

$$\text{Since } (n^8 + 1)(n^8 - 1) = n^{16} - 1$$

$$\text{By Fermat theorem, } n^{16} \equiv 1 \pmod{17}$$

$$\Rightarrow \text{One of the factors must be divisible by 17}$$

- 9 (a) Let p be a prime not equal to 3. Prove that the number $111\dots 11$ (p ones) is not divisible by p .
(b) Let $p > 5$ be a prime. Prove that the number 1111.11 ($p-1$ ones) is divisible by p .

Solution:

(a) the number $111\dots 11$ (p ones) is equal to $(10^p - 1)/9$. But $10^p - 1$ is not divisible by p , since $10^p - 1 \equiv 10 - 1 = 9 \pmod{p}$.

(b) The number $111 \dots 11$ ($p-1$ ones) is equal to $(10^{p-1}-1)/9$. But $10^{p-1}-1$ is divisible by p , since p is relatively prime to both 10 and 9.

- 10** Prove that for each prime p the difference $111\dots 11222\dots 2233\dots 33\dots 88\dots 88999\dots 99 - 123456789$ (in the first number each non-zero is written exactly p times) is divisible by p .

Solution:

For $p = 2$ and $p = 3$, this can be easily proved.

For $p > 3$

$$111 \dots 11222 \dots 2233 \dots 33 \dots 88 \dots 88999 \dots 99 = \frac{1}{9}(10^p - 1)(10^{8p} + 2 \times 10^{7p} + \dots + 9)$$

By Fermat theorem: $10^p \equiv 10 \pmod{p}$, $10^{2p} \equiv 10^2 \pmod{p}$, $10^{8p} \equiv 10^8 \pmod{p}$

$$(10^p - 1)(10^{8p} + 2 \times 10^{7p} + \dots + 9) \equiv (10^p - 1)(10^8 + 2 \times 10^7 + \dots + 9) \pmod{p}$$

$$\equiv (10^p - 1)(123456789) \pmod{p}$$

$$9(111 \dots 11222 \dots 2233 \dots 33 \dots 88 \dots 88999 \dots 99 - 123456789) \equiv (10^p - 10)(123456789) \pmod{p}$$

$$\equiv 0 \pmod{p} \quad (\text{by Fermat theorem})$$

As p is co-prime to 9 therefore given number is divisible by p .

- 11** Show that, $11 \mid (5^{10} - 3^{10})$. More generally, if p is a prime such that $p \nmid a$ and $p \nmid b$, then show that $p \mid (a^{p-1} - b^{p-1})$.

Solution:

$$5^{10} - 3^{10} \equiv 1 - 1 \equiv 0 \pmod{11} \quad (\text{By Fermat theorem})$$

$$\text{Similarly, } a^{p-1} - b^{p-1} \equiv 1 - 1 \equiv 0 \pmod{p}$$

- 12** Prove that and (i) $7 \mid (111^{333} + 333^{111})$ (ii) $39 \mid (53^{103} + 103^{53})$

Solution:

$$(i) 111^{333} + 333^{111} \equiv (-1)^{333} + (4)^{111} \pmod{7}$$

$$4^3 \equiv 1 \pmod{7} \Rightarrow 4^{111} \equiv 1 \pmod{7}$$

$$111^{333} + 333^{111} \equiv -1 + 1 \equiv 0 \pmod{7}$$

$$(ii) 53^{103} + 103^{53} \equiv (2)^{103} + (1)^{53} \pmod{3}$$

$$\equiv (4)^{51} \times 2 + 1 \pmod{3}$$

$$\equiv (1)^{51} \times 2 + 1 \equiv 0 \pmod{3}$$

$$53^{103} + 103^{53} \equiv (1)^{103} + (-1)^{53} \pmod{13}$$

$$\equiv 1 - 1 \equiv 0 \pmod{13}$$

$$\Rightarrow 53^{103} + 103^{53} \text{ divisible by } 39.$$

- 13** If p and q are distinct primes, show that $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$

Solution:

$$p^{q-1} + q^{p-1} \equiv 1+0 \pmod{q} \text{ \& } p^{q-1} + q^{p-1} \equiv 0+1 \pmod{p}$$

$$p^{q-1} + q^{p-1} = kq + 1 \equiv 1 \pmod{p}, k \in \mathbb{Z}$$

$$\Rightarrow k = mp$$

$$\Rightarrow p^{q-1} + q^{p-1} = 1 + mpq, k \in \mathbb{Z}$$

$$\Rightarrow p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$$

- 14** Find the remainder when $7^{200} + 11^{800}$ is divided by 101

Solution:

$$7^{100} \equiv 1 \pmod{101} \text{ and } 11^{100} \equiv 1 \pmod{101}$$

$$7^{200} \equiv 1 \pmod{101} \text{ and } 11^{800} \equiv 1 \pmod{101}$$

$$7^{200} + 11^{800} \equiv 2 \pmod{101}$$

- 15** Show that $89 \mid (2^{44} - 1)$ and $97 \mid (2^{48} - 1)$

Solution:

$$2^6 \equiv -25 \pmod{89}$$

$$\Rightarrow 2^{12} \equiv 625 \equiv 2 \pmod{89}$$

$$\Rightarrow 2^{11} \equiv 1 \pmod{89}$$

$$\Rightarrow 2^{44} \equiv 1 \pmod{89}$$

$$2^6 \equiv -33 \pmod{97}$$

$$\Rightarrow 2^{12} \equiv 1089 \equiv 22 \pmod{97}$$

$$\Rightarrow 2^{24} \equiv 484 \equiv -1 \pmod{97}$$

$$\Rightarrow 2^{48} \equiv 1 \pmod{97}$$